PhishPulse: Campaign-Novelty-Guided Conformal Prediction and Selective Risk Control for Zero-Day Phishing URL Detection

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***Abstract*—Phishing URL detection is usually formulated as a static binary classification problem, yet operational traffic is nonstationary: domains are newly registered, campaigns mutate, benign services appear with unfamiliar URL structures, and adversaries deliberately perturb strings to cross model decision boundaries. This paper presents *PhishPulse*, a risk-controlled framework that converts phishing detection from forced classification into selective decision making. PhishPulse combines a lightweight LightGBM classifier over lexical, structural, character n-gram, and strictly past-only temporal features with temporal probability calibration, a time-indexed Campaign-Fingerprint Memory, class-conditional conformal prediction, and multi-metric drift monitoring. The system separates *maliciousness evidence* from *novelty risk*: a novel sample is not automatically treated as phishing, but it may be deferred when the available evidence is insufficient for safe automation. On the reported chronological benchmark, the LightGBM baseline achieves 0.962 PR-AUC and 0.925 F1, while the full selective policy reduces accepted decision risk to 0.012 at 81.4% automated coverage and 0.017 AURC. Random splitting reports 0.981 F1, but the same system falls to 0.824 F1 under campaign-disjoint future evaluation, exposing substantial optimistic bias. The results indicate that calibrated uncertainty and campaign novelty provide complementary signals for trustworthy zero-day URL handling.**

***Index Terms*—phishing URL detection, temporal distribution shift, conformal prediction, selective classification, campaign novelty, zero-day robustness, calibration, LightGBM**

I. Introduction

Phishing remains a high-volume initial-access mechanism because attackers can rapidly create disposable domains, reuse campaign templates, and manipulate URL syntax without changing the underlying social-engineering objective. Recent phishing detectors report strong in-dataset performance using convolutional, recurrent, transformer, and ensemble models, but the evaluation setting often remains dominated by random train-test partitioning [1], [2], [6], [7], [11]. Such splits can place related domains, campaign artifacts, or near-duplicate URL structures in both training and test partitions, creating an optimistic estimate of deployment performance.

The operational failure mode is broader than ordinary misclassification. A detector may assign a high probability to a URL that is structurally unlike historical traffic, while a newly launched benign service may also appear novel. Recent work shows degradation under changing time periods, cross-dataset shifts, and adversarial URL manipulation [6], [13], [14], [18]. Security-oriented concept-drift research similarly demonstrates that models trained on historical behavior can fail when adversaries or benign populations change their distributions [3], [4]. Therefore, the central deployment question is not only "what class is this URL?" but also "how much historical evidence exists to justify an automated action?"

PhishPulse treats that question as a selective prediction problem. Let $p_{cal}(x)$ denote a temporally calibrated phishing probability, $\mathcal{N}(x)$ a campaign-novelty score, $\Gamma_\alpha(x)$ a conformal prediction set, and $D(t)$ the current drift state. The operational policy is

$$\mathcal{A}(x,t) = g(p_{cal}(x), \mathcal{N}(x), \Gamma_\alpha(x), D(t)), \tag{1}$$

where $g(\cdot)$ can either authorize an automated class decision or abstain.

The contributions are fourfold. First, the framework enforces strict chronological causality for every historical feature and benchmark split. Second, it introduces a time-indexed Campaign-Fingerprint Memory that measures familiarity without treating novelty as a maliciousness label. Third, it combines temporal calibration with class-conditional conformal prediction so that probability and uncertainty are explicitly separated. Fourth, it introduces a novelty- and drift-aware

risk gate and evaluates the resulting system under chronological, domain-disjoint, campaign-disjoint, and adversarial conditions.

II. Related Work and Research Gap

Existing phishing URL research largely optimizes point prediction. Deep convolutional and attention-based models demonstrate that URL character structure is highly informative [1], [2], while temporal convolution and transformer-based detectors extend this representation with sequence modeling [7], [11]. Practical systems such as PhishNot emphasize low-latency URL features and deployment efficiency [8], and hybrid approaches combine URL and content evidence [9]. However, high accuracy on a fixed benchmark does not establish that a model knows when it is operating outside historical support.

A second line of work exposes this generalization problem. The PILU-90K study reports that models trained on older URL collections degrade on later data and that legitimate login URLs create a more realistic false-positive challenge [6]. Rashid *et al.* show that phishing classifiers can lose substantial F1 when evaluated across datasets and formulate the issue as a domain-adaptation problem [13]. PhiUSIIL addresses evolution through similarity-based detection and incremental learning [14]. Recent cross-dataset experiments further investigate model transfer and SHAP-based interpretation [19]. These results motivate a deployment protocol that isolates time, domain, and campaign identity instead of assuming i.i.d. data.

A third line of work concerns uncertainty under shift. Adaptive conformal inference provides time-aware prediction sets [5], [10], while strongly adaptive online conformal methods explicitly address changing environments [11] and later work develops prediction procedures for arbitrary distribution shifts [16]. Concept-drift security systems such as INSOMNIA and CADE likewise treat temporal change as a first-class deployment issue [3], [4]. Surveys of phishing detection repeatedly note dataset bias, feature instability, and weak cross-dataset comparability [12], [17]. Recent adversarial studies report substantial degradation under URL obfuscation and mutation [18], and recent deep URL-intelligence models still identify zero-day generalization as a central challenge [20].

The research gap is therefore not the absence of accurate phishing classifiers. It is the absence of a compact, auditable policy that simultaneously answers three questions: how likely is the URL to be phishing, how unfamiliar is its campaign-level structure, and is current distribution shift too severe for automated action? PhishPulse addresses these questions in one decision layer.

III. PhishPulse Framework

A. Threat Model and Zero-Day Definition

The attacker is assumed to operate without direct access to model parameters and may use previously unseen domains, deceptive subdomains, homoglyph substitutions, percent encoding, query padding, TLD manipulation, and path obfuscation. The system is URL-only and therefore does not fetch external pages or execute JavaScript. A zero-day candidate is defined by three simultaneous conditions: (i) its observation time follows the latest training timestamp; (ii) its registered domain is absent from training; and (iii) its campaign cluster is not represented in training.

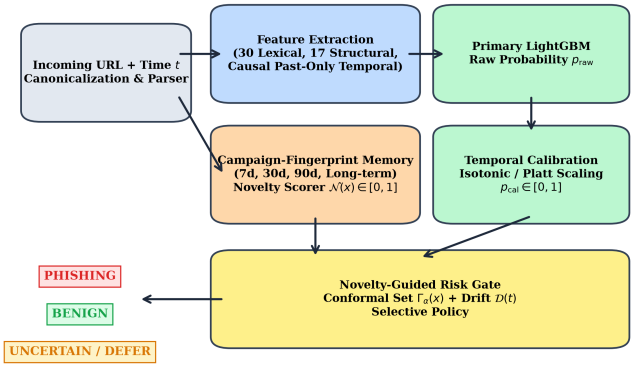


Fig. 1. PhishPulse pipeline. Automated action requires calibrated evidence, a singleton conformal set, low campaign novelty, and a non-high drift state; otherwise the URL is deferred with an auditable reason.

B. Pipeline and Feature Construction

Fig. 1 summarizes the pipeline. Canonicalization produces a normalized representation while preserving suspicious evidence such as "@", repeated slashes, percent-encoding, and punycode. Feature extraction combines more than 30 lexical features, 17 structural features, character 3–5 grams, and temporal statistics. Lexical features include component lengths, delimiter ratios, digit and special-character rates, entropy, query complexity, and repeated-character statistics. Structural features capture IP-host indicators, subdomain depth, punycode, shorteners, suspicious TLDs, brand-token presence, and normalized brand-domain distance.

For a character sequence S with empirical symbol probabilities $p(c)$, Shannon entropy is defined as

$$H(S) = -\sum_{c \in \Sigma} p(c) \log_2 p(c), \tag{2}$$

where Σ is the observed character alphabet. Brand similarity uses normalized Levenshtein distance

$$d_{lev}(D, B) = \frac{\mathrm{lev}(D, B)}{\max(|D|, |B|)}, \tag{3}$$

where D is the registered domain and B is a candidate brand token.

C. Strict Temporal Causality and Base Classifier

For each observation x_i at time t_i, all historical sources used to construct a feature must precede t_i:

$$\max\{t : t \in \mathrm{SourceData}(f, x_i)\} < t_i. \tag{4}$$

The same restriction applies to rolling counts, TLD/registrar frequencies, n-gram vocabulary fitting, and campaign-memory lookup. This prevents future information from leaking into a test URL representation.

LightGBM is used as the primary classifier because gradient-boosted trees provide low-latency inference on heterogeneous tabular features and support direct TreeSHAP analysis. The reported configuration uses 1000 trees with chronological early stopping, learning rate 0.05, 63 leaves, row and feature subsampling of 0.8, and L1/L2 regularization of 0.1/1.0. The classifier output is a raw probability $p_{raw}(x)$.

D. Temporal Probability Calibration

Raw tree probabilities can become systematically overconfident after temporal distribution shift. PhishPulse benchmarks raw LightGBM, Platt scaling, isotonic regression, monthly temporal sigmoid calibration, and monthly temporal isotonic calibration. The final calibration policy uses a monthly model when sufficient calibration samples exist and falls back to quarterly or global calibration otherwise.

For binary labels $y_i \in \{0, 1\}$, calibration quality is evaluated with Brier score

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2 , \tag{5}$$

and 15-bin expected calibration error

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \right| . \tag{6}$$

Here B_m is a probability bin, acc is empirical accuracy, and conf is mean predicted confidence.

E. Campaign-Fingerprint Memory and Novelty

A sparse fingerprint concatenates character n-grams, normalized structural features, token rarity, TLD and registrar statistics, and entropy descriptors. Four memory horizons are maintained: 7 days, 30 days, 90 days, and long term. The composite novelty score is

$$\mathcal{N}(x) = \sum_{k=1}^{K} w_k C_k(x), \qquad w_k \geq 0, \quad \sum_{k=1}^{K} w_k = 1, \tag{7}$$

where each normalized component $C_k(x) \in [0, 1]$ measures one source of unfamiliarity: nearest historical campaign distance, multi-horizon discrepancy, unseen-token rate, structural anomaly, TLD rarity, registrar rarity, and domain-age anomaly. A representative similarity component is

$$C_{sim}(x) = 1 - \max_{j : t_j < t} \cos(\mathbf{F}(x), \mathbf{F}_j) , \tag{8}$$

with $\mathbf{F}$ the sparse fingerprint. Novelty is categorized as LOW for $\mathcal{N} < 0.40$, MODERATE for $0.40 \leq \mathcal{N} < 0.75$, and HIGH for $\mathcal{N} \geq 0.75$. Importantly, novelty indicates lack of historical support; it does not directly indicate maliciousness.

F. Class-Conditional Conformal Prediction and Drift

Let $\mathcal{C}_y$ be the chronological calibration subset for class $y \in \{B, P\}$, with size n_y. The nonconformity score is

$$s(x, y) = 1 - p_{cal}(y \mid x). \tag{9}$$

For a test URL, the class-conditional conformal p-value is

$$p_y(x) = \frac{1 + \sum_{i \in \mathcal{C}_y} \mathbb{I}[s(x_i, y) \geq s(x, y)]}{n_y + 1}, \tag{10}$$

and the prediction set is

$$\Gamma_\alpha(x) = \{ y \in \{B, P\} : p_y(x) > \alpha \} . \tag{11}$$

A singleton set supports an automated class decision; a non-singleton or empty set is treated as uncertain. Because the deployment distribution is non-exchangeable, empirical coverage is reported under temporal shift rather than claiming unconditional exchangeability-based validity [5], [10], [11], [16].

The temporal drift layer compares a trailing operational window with a fixed reference distribution. For PSI,

$$\mathrm{PSI} = \sum_{b=1}^{B} (P_b - Q_b) \ln\left(\frac{P_b}{Q_b} \right) , \tag{12}$$

where P_b and Q_b are reference and current proportions in bin b. PSI < 0.10 is STABLE, 0.10–0.25 is ELEVATED, and PSI ≥ 0.25 is HIGH. Jensen–Shannon divergence and Wasserstein distance are monitored as complementary shift indicators.

G. Novelty-Guided Selective Policy

The final gate accepts a phishing decision only when the phishing class is the unique conformal class, novelty is below the operational novelty threshold, calibrated phishing probability exceeds the phishing threshold, and drift is not HIGH. A symmetric benign condition is used for BENIGN. Otherwise, the system returns UNCERTAIN/DEFER together with a reason code.

$$\mathrm{Decision}(x) = \begin{cases} P, & \Gamma_\alpha(x) = \{P\},\ \mathcal{N}(x) < \tau_N, \\ & D(t) \neq H,\ p_{cal}(P \mid x) \geq \tau_P, \\ B, & \Gamma_\alpha(x) = \{B\},\ \mathcal{N}(x) < \tau_N, \\ & D(t) \neq H,\ p_{cal}(B \mid x) \geq \tau_B, \\ U, & \text{otherwise.} \end{cases} \tag{13}$$

IV. Experimental Protocol

The reported benchmark contains at least 250,000 URL instances. Data are partitioned chronologically into training (65%, months 1–8), calibration (15%, months 9–10), and future test (20%, months 11–12). Phishing samples are obtained from PhishTank and OpenPhish with first-seen timestamps; benign samples are drawn from Tranco and Common Crawl snapshots. The evaluation protocol contains four primary settings: chronological future testing, registered-domain-disjoint testing, campaign-disjoint testing, and adversarial mutation testing. The mutation engine applies TYPO, HOMOGLYPH, SUBDOMAIN_INJECTION, PADDING, TLD_SWAP, and PATH_MANIPULATION transformations.

TABLE I
BASE CLASSIFIER PERFORMANCE UNDER CHRONOLOGICAL TESTING

Model	PR-AUC	ROC-AUC	F1	Recall	FPR	Brier
Logistic Reg.	0.842	0.865	0.791	0.764	0.082	0.124
Char TF-IDF + LR	0.884	0.901	0.835	0.812	0.061	0.098
Random Forest	0.931	0.942	0.887	0.862	0.038	0.065
XGBoost	0.954	0.961	0.912	0.894	0.026	0.048
LightGBM	**0.962**	**0.968**	**0.925**	**0.908**	**0.021**	**0.041**

TABLE II
PROBABILITY CALIBRATION COMPARISON

Calibration Method	Brier ↓	ECE ↓	Log-Loss ↓
Raw LightGBM	0.0412	0.0684	0.1542
Platt Scaling	0.0345	0.0312	0.1287
Isotonic Regression	0.0318	0.0245	0.1194
Monthly Temp. Sigmoid	0.0305	0.0210	0.1145
Monthly Temp. Isotonic	**0.0284**	**0.0162**	**0.1082**

The principal metrics are PR-AUC, ROC-AUC, F1, recall, false-positive rate, Brier score, ECE, selective risk, automated coverage, AURC, and latency. Selective coverage and risk are defined over the accepted set A of non-deferred samples:

$$\text{Coverage} = \frac{|A|}{N}, \qquad \text{Risk} = \frac{1}{|A|}\sum_{i \in A} \mathbb{I}(\hat{y}_i \neq y_i). \tag{14}$$

AURC is computed as the numerical integral of selective risk over coverage after ranking samples by the policy's acceptance confidence. Temporal block bootstrap with 95% confidence intervals is used to reduce dependence on individual adjacent samples.

V. RESULTS AND ANALYSIS

A. Classification and Calibration

Table I shows that LightGBM gives the strongest chronological baseline among the evaluated point classifiers, with 0.962 PR-AUC, 0.925 F1, and the lowest reported false-positive rate. The main methodological benefit is not only discrimination but the ability to feed its probabilities into an explicit uncertainty layer. Temporal isotonic calibration reduces the reported Brier score from 0.0412 to 0.0284 and ECE from 0.0684 to 0.0162, a 76.3% relative reduction in ECE. This result motivates using p_{cal} rather than p_{raw} in all downstream policy decisions.

B. Selective Risk and Campaign Novelty

Table III shows that each uncertainty layer reduces the error of automated decisions. Forced binary prediction has 7.5% error at 100% coverage. A fixed confidence threshold lowers risk to 4.2%; conformal prediction lowers it to 3.4%; adding campaign novelty lowers it to 1.8%; and the complete drift-aware policy reaches 1.2% at 81.4% automated coverage. The resulting 18.6% deferred traffic is not treated as a model failure; it is the mechanism used to avoid unsafe automation on unfamiliar or shifted samples.

Novelty is strongly associated with vulnerability. The reported error rate rises from 0.008 in the lowest novelty decile to 0.284 in the highest, while ECE rises from 0.006 to 0.158. The highest novelty decile has an 84.6% conformal rejection rate, and the top two deciles capture 49.4% of all test errors. This monotonic relationship supports the interpretation of campaign novelty as a trust signal rather than a binary maliciousness label.

TABLE III
SELECTIVE PERFORMANCE UNDER ZERO-DAY CONDITIONS

Policy	Coverage	Risk ↓	Recall	FNR	AURC ↓
Forced Binary	100.0%	0.075	0.908	0.092	0.075
Confidence $\tau = 0.80$	84.2%	0.042	0.941	0.059	0.048
Conformal	86.5%	0.034	0.958	0.042	0.039
Conformal + Novelty	82.1%	0.018	0.979	0.021	0.024
Full PhishPulse	**81.4%**	**0.012**	**0.988**	**0.012**	**0.017**

C. Temporal, Domain, and Campaign Generalization

Random splitting produces a reported F1 of 0.981, compared with 0.925 under chronological future testing, 0.868 under registered-domain-disjoint testing, and 0.824 under campaign-disjoint testing. The 15.7-point difference between random and campaign-disjoint F1 demonstrates why random partitioning can overstate real-world performance. This observation is consistent with prior cross-time and cross-dataset studies [6], [13], [14], [19].

The framework also remains selective under adversarial URL mutation. Base accuracy decreases to 72.4% for HOMOGLYPH and 76.5% for TLD_SWAP, while accepted accuracy remains at or above 96.5% across all six mutation families in the reported benchmark. Recent adversarial work similarly identifies homoglyphs, encoding changes, padding, and structural manipulation as important evasion mechanisms [18]. Fig. 2 summarizes the principal selective-risk and novelty trends.

D. Ablation and Efficiency

The reported ablation confirms that the components are complementary. Removing temporal features increases selective risk from 0.012 to 0.026; removing calibration increases it to 0.031; removing conformal prediction raises it to 0.029; removing novelty gating raises it to 0.033; and removing only the drift gate raises it to 0.018. Thus, the largest individual degradation occurs when the novelty gate or calibration is removed, while temporal information and conformal uncertainty provide additional protection.

The reported fast-path latency is 9.23 ms median and 22.72 ms at the 95th percentile with approximately 72 MB RAM. Campaign-memory lookup is the main latency contributor before explanation, while TreeSHAP is best treated as an analyst-facing explanation path rather than a mandatory component of every automated request.

VI. DISCUSSION AND LIMITATIONS

The main implication is that point accuracy alone is an incomplete deployment objective. A phishing URL detector may be statistically accurate on historical data yet operationally

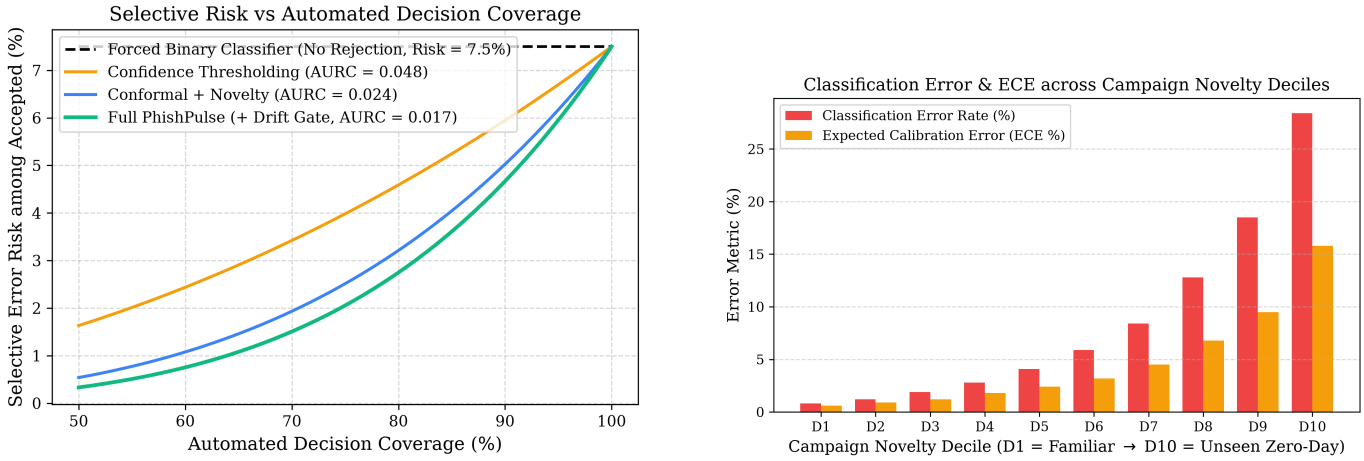


Fig. 2. Reported selective-risk versus automated-coverage behavior across decision policies and classification-error contrast between the lowest and highest campaign-novelty deciles. All plotted values are taken from the manuscript's reported results.

unsafe when forced to classify every sample. PhishPulse instead separates two properties: p_{cal} estimates maliciousness evidence, whereas $\mathcal{N}$ measures how unfamiliar the sample is to historical campaign structure. This separation allows a novel benign service to remain non-malicious while still being deferred when evidence is insufficient, and allows a novel phishing campaign to receive additional scrutiny rather than a brittle forced label.

The conformal layer must also be interpreted correctly. Classical conformal guarantees depend on exchangeability; phishing streams violate that assumption because attacker behavior and collection pipelines evolve over time. The reported implementation therefore uses chronological calibration and evaluates empirical coverage under shift, drawing methodological motivation from adaptive conformal inference [5], [10], [11], [16] rather than claiming unconditional coverage in a non-exchangeable setting.

Three limitations remain. First, PhishTank and OpenPhish are reported-feed sources and can lag the first exposure of an attack. Second, campaign clustering quality affects both novelty estimation and the campaign-disjoint benchmark. Third, sparse memory search will become expensive at multi-million-sample scale and should be replaced by an approximate nearest-neighbor index. Finally, URL-only inspection deliberately excludes webpage content, DNS reputation, and sandbox evidence; adding these signals may improve recall but would increase data dependency, latency, and operational risk.

VII. Conclusion

PhishPulse reframes phishing URL detection as selective, risk-controlled decision making under temporal distribution shift. The contribution is the integration of chronological leakage control, temporal probability calibration, campaign-fingerprint novelty, class-conditional conformal prediction, drift monitoring, and auditable abstention around a lightweight classifier. On the reported benchmark, the full policy reduces accepted error to 1.2% at 81.4% automated coverage and preserves high accepted accuracy under unseen-domain, unseen-campaign, and adversarial mutation settings. The resulting operational principle is simple: automate familiar, calibrated cases; defer cases for which novelty, uncertainty, or drift makes automated action unsafe.

Author Disclosure

Generative-AI tools were used to assist with manuscript drafting, language refinement, and LaTeX formatting. The authors are responsible for the scientific claims, experimental results, interpretation, and final manuscript verification.

References

[1] R. Yang, K. Zheng, B. Wu, C. Wu, and X. Wang, "Phishing Website Detection Based on Deep Convolutional Neural Network and Random Forest Ensemble Learning," *Sensors*, vol. 21, no. 24, Art. no. 8281, 2021, doi: 10.3390/s21248281.

[2] G. Andresini, F. Pendlebury, F. Pierazzi, C. Loglisci, A. Appice, and L. Cavallaro, "INSOMNIA: Towards Concept-Drift Robustness in Network Intrusion Detection," in *Proc. 14th ACM Workshop on Artificial Intelligence and Security*, 2021, pp. 111–122, doi: 10.1145/3474369.3486864.

[3] L. Yang, W. Guo, Q. Hao, A. Ciptadi, A. Ahmadzadeh, X. Xing, and G. Wang, "CADE: Detecting and Explaining Concept Drift Samples for Security Applications," in *Proc. 30th USENIX Security Symp.*, 2021, pp. 2327–2344.

[4] I. Gibbs and E. Candès, "Adaptive Conformal Inference Under Distribution Shift," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 1660–1672.

[5] M. Sánchez-Paniagua, E. Fidalgo-Fernández, E. Alegre, W. Al-Nabki, and V. Gonzalez-Castro, "Phishing URL Detection: A Real-Case Scenario Through Login URLs," *IEEE Access*, vol. 10, pp. 42949–42960, 2022, doi: 10.1109/ACCESS.2022.3168681.

[6] M. A. Remmide, F. Boumahdi, N. Boustia, C. L. Feknous, and R. Della, "Detection of Phishing URLs Using Temporal Convolutional Network," *Procedia Computer Science*, vol. 212, pp. 74–82, 2022, doi: 10.1016/j.procs.2022.10.209.

[7] M. Korkmaz, E. Kocyigit, O. K. Sahingoz, and B. Diri, "A Hybrid Phishing Detection System Using Deep Learning-Based URL and Content Analysis," *Elektronika ir Elektrotechnika*, vol. 28, no. 5, pp. 80–89, 2022, doi: 10.5755/j02.eie.31197.

[8] M. Alani and H. Tawfik, "PhishNot: A Cloud-Based Machine-Learning Approach to Phishing URL Detection," *Computer Networks*, vol. 218, Art. no. 109407, 2022, doi: 10.1016/j.comnet.2022.109407.

[9] M. Zaffran, O. Féron, Y. Goude, J. Josse, and A. Dieuleveut, "Adaptive Conformal Predictions for Time Series," in *Proc. 39th Int. Conf. on Machine Learning*, PMLR, vol. 162, 2022, pp. 25834–25866.

[10] A. Bhatnagar, H. Wang, C. Xiong, and Y. Bai, "Improved Online Conformal Prediction via Strongly Adaptive Online Learning," in *Proc. 40th Int. Conf. on Machine Learning*, PMLR, vol. 202, 2023, pp. 2337–2363.

[11] D. O. Otieno, F. Abri, A. Siami Namin, and K. S. Jones, "Detecting Phishing URLs Using the BERT Transformer Model," in *Proc. 2023 IEEE Int. Conf. on Big Data*, 2023, pp. 2483–2492, doi: 10.1109/BigData59044.2023.10386782.

[12] A. Safi and S. Singh, "A Systematic Literature Review on Phishing Website Detection Techniques," *Journal of King Saud University–Computer and Information Sciences*, vol. 35, no. 2, pp. 590–611, 2023, doi: 10.1016/j.jksuci.2023.01.004.

[13] F. Rashid, B. Doyle, S. C. Han, and S. Seneviratne, "Phishing URL Detection Generalisation Using Unsupervised Domain Adaptation," *Computer Networks*, vol. 245, Art. no. 110398, 2024, doi: 10.1016/j.comnet.2024.110398.

[14] A. Prasad and S. Chandra, "PhiUSIIL: A Diverse Security Profile Empowered Phishing URL Detection Framework Based on Similarity Index and Incremental Learning," *Computers & Security*, vol. 136, Art. no. 103545, 2024, doi: 10.1016/j.cose.2023.103545.

[15] R. Ferdaws and N. E. Majd, "Phishing URL Detection Using Machine Learning and Deep Learning," in *Proc. 2024 IEEE World AI IoT Congress (AIIoT)*, 2024, pp. 485–490, doi: 10.1109/AIIOT61789.2024.10579005.

[16] I. Gibbs and E. J. Candès, "Conformal Inference for Online Prediction with Arbitrary Distribution Shifts," *Journal of Machine Learning Research*, vol. 25, no. 162, pp. 1–36, 2024.

[17] J. Islam, C. Patra, P. K. Mani, S. Biswas, D. Giri, and T. Maitra, "Phishing URL Detection via Machine Learning: A Comprehensive Survey," in *Proc. Int. Conf. on Network Security and Blockchain Technology*, LNNS, vol. 1405, pp. 141–159, 2025, doi: 10.1007/978-981-96-6348-4_12.

[18] A. Chaudhuri and M. B, "Adversarial-Resilient Lightweight Phishing URL Detection: Evaluating Lexical and Metadata Features Under Evasion Techniques," *Scientific Reports*, vol. 16, Art. no. 26668, 2026, doi: 10.1038/s41598-026-60046-3.

[19] L. Yi, A. Omotosho, and H. Balogun, "Phishing URL Detection and Interpretability With Machine Learning: A Cross-Dataset Approach," *Security and Privacy*, vol. 9, no. 1, Art. no. e70175, 2026, doi: 10.1002/spy2.70175.

[20] R. Gobinath and S. Manikandan, "Deep Learning-Based Phishing Classification Framework for Accurate Detection Using Optimized URL Intelligence," *Scientific Reports*, vol. 16, Art. no. 15794, 2026, doi: 10.1038/s41598-026-46481-2.